TASK 1: READING — 2 HOURS BEFORE CLASS

Explore evolutionary algorithm parameter management (initial tuning and continuous control) by reading and reflecting on chapters 7 and 8 of the following book:

A. E. Eiben and James E. Smith. 2015. Introduction to Evolutionary Computing (2nd. ed.). Springer Publishing Company, Incorporated - full text available at https://read.kortext.com/reader/pdf/406661/C1

As you read, make notes on the core concepts:

1. What are evolutionary algorithm parameters? Give examples.
2. How are parameters used to design and implement distinct evolutionary algorithm instances?
3. What is parameter tuning and when does it occur?
4. What is the relationship between parameter tuning and algorithm quality (performance and robustness)?
5. What are the traditional, most effective parameter tuning methods and how do they lead to higher algorithm quality?
6. What is parameter control and when does it occur? Use examples to illustrate: mutation step size and penalty coefficients.
7. What are the main types of parameter control relative to: (a) the component of the algorithm being controlled; (b) the manner in which the control is enforced; and (c) the knowledge informing the control process?
8. What are the traditional parameter control techniques applicable to: (a) representation; (b) evaluation; (c) recombination; (d) selection; and (e) population size?
9. What are the key insights to be gained from traditional parameter management?

TASK 2: DISCUSSION — 35 MIN DURING CLASS

As a group, discuss your individual notes and those of your peers. Produce a summary of the material covered in the pre-class reading, incorporating all 9 points above. — 15 MIN

Students will present the 9 points one by one, in 2 minutes each. — 20 MIN

TASK 3: REFLECTION — 15 MIN DURING CLASS

Amend your summary based on the discussion. Turn the revised summary into a set of presentation slides that you can build upon for the related part of your coursework presentation.

RELEVANCE TO MODULE ASSESSMENT

Completing the pre-class reading and reflection, taking notes, revising and collating them in a group to form a summary, amending the summary with insights gathered from group discussion, and preparing and delivering a presentation underpinned by the updated summary will positively influence the marks you will be awarded for the element of the coursework relevant to parameter management.